Microwave Spectroscopy

ROMY

Questions related with class lecture

01. Define 'Microwave spectroscopy' & 'Moment of inertia'

Microwave spectroscopy:

Spectroscopy in the microwave region of the electromagnetic radiation is known as microwave spectroscopy. In other words microwave spectroscopy deals with the interaction of electromagnetic radiation with a rotating molecule in the microwave region of the electromagnetic radiation.

Moment of Inertia: [2010]

Matter is composed of some particles. The sum of the mass of each particle & their square distance from rotational axis with respect to the rotational axis is known as the moment of inertia of that matter. It is denoted by 'I'.

If 'm' be the mass of each particle & 'r' be the square distance from the rotational axis, then-

$$I = \sum mr^2$$

Any body has three principle moment of inertia. They are usually designated by-

P.T.O.

ⓐ (I_A) about the head axis.

ⓑ (I_B) end-over-end rotation in the plane of paper.

ⓒ (I_C) end-over-end rotation at the right angle to the plane.

02. Write shortly about the 'Rotation of molecules' or 'Rotating molecules'. Discuss briefly about the classification of rotating molecules according to the three principle moment of inertia.

⇨ Rotation of molecules:

Spectroscopy in the microwave region is concerned with the study of rotating molecules. The rotation of a three-dimensional body may be quite complex & it is convenient to resolve it into the rotational components about three mutually perpendicular directions through the center of gravity - the principal axis of rotation.

Classification of rotating molecules:

A body has three principle moments of inertia, one about each axis, usually designated by I_A, I_B & I_C. Molecules may be classified into the following groups according to the relative values of their principal moments of inertia. The classification of rotating molecules are given below-

(I) Linear molecules,

(II) Symmetric top molecules,

(III) Spherical top molecules &

(IV) Asymmetric top molecules.

Among them, spherical top molecules are microwave inactive. All rotating molecules are described below.

(I) Linear molecules:

The molecules in which all the atoms are arranged in a straight line, such as Hydrogen chloride (HCl) or, carbon oxysulphide (OCS) illustrated below—

$$H-Cl \quad ; \quad O=C=S.$$

The three directions of rotation may be taken as- (a) about the bond axis, (b) end-over-end rotation in the plane of the paper, & (c) end-over-end rotation at right angles to the plane. Here, the moments of (b) & (c) are the same (that is $I_B = I_C$) while that of (a) is very small. As an approximation we may say that $I_A = 0$. Thus for linear molecules we have-

$$I_B = I_C \quad ; \quad I_A = 0.$$

(II) Symmetric top molecules:

Consider a molecule such as methyl fluoride, where the three hydrogen atoms are bonded tetrahedrally

P.T.O.

⇨ Rotational spectra: [2009]

We know that the rotational energy of a molecule is quantized, i.e. a molecule can not have any arbitrary (যেকোনো) amount of rotational energy. Thus transition from one rotational energy level to another rotational energy level involves finite amount of energy transfer which may result a rotational spectrum.

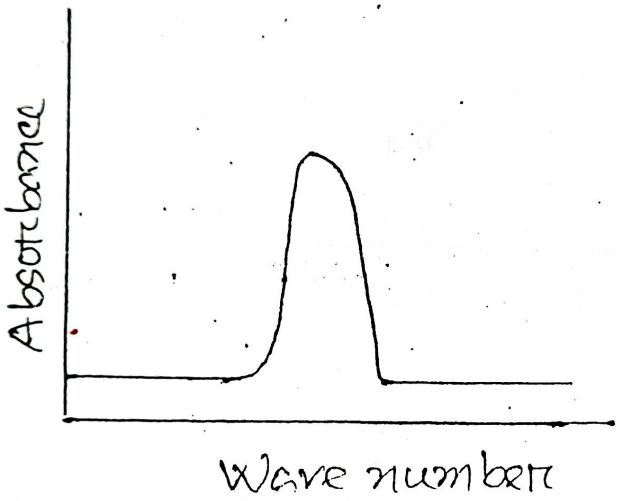


Figure: Rotational spectrum of a molecule for single transition only.

Expression for the rotational spectrum of rigid di-atomic molecules: (L) ✓

Consider the case of a rigid diatomic molecule consisting of two atoms of masses m_1 & m_2 as shown in the figure.

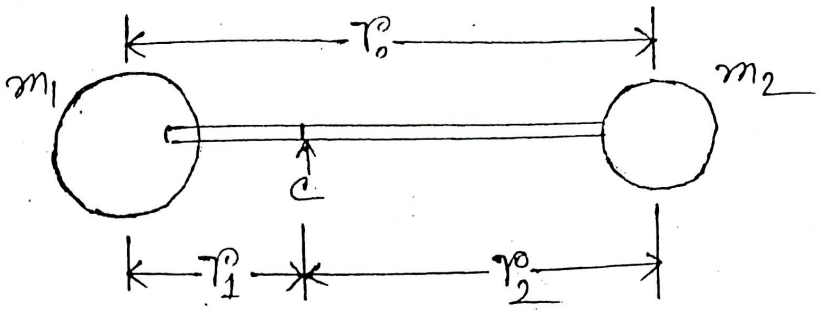


Figure: The hypothetical rigid diatomic molecule.

The molecule can be considered as two masses m_1 & m_2 joined by a rigid bar (the bond) whose length is —

$$r_0 = r_1 + r_2 \longrightarrow (I)$$

Where r_1 & r_2 are the distances of m_1 & m_2 from the centre of gravity C respectively.

If the molecule rotates end-over-end about the point C, then →

$$m_1 r_1 = m_2 r_2 \longrightarrow (II)$$

The total moment of inertia about C is given by—

$$I = m_1 r_1^2 + m_2 r_2^2 \longrightarrow (III)$$

$$\Rightarrow I = m_1 r_1 \cdot r_1 + m_2 r_2 \cdot r_2$$

$$\Rightarrow I = m_2 r_2 r_1 + m_1 r_1 r_2 \quad [\because m_1 r_1 = m_2 r_2]$$

$$\therefore I = r_1 r_2 (m_1 + m_2) \longrightarrow (IV)$$

Again from eqn. (III)

$$I = m_1 r_1^2 + m_2 r_2^2$$

$$\Rightarrow I = m_2 r_2 r_1 + m_2 r_2^2$$

P.T.O.

$\Rightarrow I = m_2 r_2 (r_1 + r_2)$

$\Rightarrow I = m_2 r_2 r_0$ [from eqn (I)]

$\therefore I = m_2 r_2 r_0 \longrightarrow (V)$

From eqns. (IV) & (V) we have,

$$r_1 r_2 (m_1 + m_2) = m_2 r_2 r_0$$

$$\Rightarrow r_1 = \frac{m_2 r_0}{m_1 + m_2} \longrightarrow (VI)$$

Now, Putting the value of r_1 in eqn. (I),

$$r_0 = r_1 + r_2$$

$$\Rightarrow r_0 - r_1 = r_2$$

$$\Rightarrow r_2 = r_0 - \frac{m_2 r_0}{m_1 + m_2}$$

$$\Rightarrow r_2 = \frac{m_1 r_0 + m_2 r_0 - m_2 r_0}{m_1 + m_2}$$

$$\Rightarrow r_2 = \frac{m_1 r_0}{m_1 + m_2} \longrightarrow (VII)$$

Now, the total moment of inertia,

$$I = r_1 r_2 (m_1 + m_2)$$

$$\Rightarrow I = \frac{m_2 r_0}{(m_1 + m_2)} \times \frac{m_1 r_0}{(m_1 + m_2)} \times (m_1 + m_2)$$

$$\Rightarrow I = \frac{m_1 m_2}{m_1 + m_2} r_0^2$$

$$\therefore I = \mu r_0^2 \longrightarrow (VIII)$$

Where, $\mu = \frac{m_1 m_2}{m_1 + m_2}$; is called the reduced mass of the system.

P.T.O

Now, by the use of Schrödinger eqn. it can be shown that the rotational energy levels allowed to the rigid diatomic molecule are given by the following expression—

$$E_J = \frac{\hbar^2}{8\pi^2 I} J(J+1) \text{ joules} \qquad \text{(IX)}$$

where,

$\hbar$ = Plancks constant

I = Moment of inertia

J = rotational quantum number.

i.e. $J = 0, 1, 2, 3 \ldots\ldots$

c = speed of light

In rotational region, spectra are usually discussed in terms of wave number, so it is useful to consider energies expressed in these units.

Thus, we have,

$$\varepsilon_J = \frac{E_J}{hc} = \frac{\hbar}{8\pi^2 Ic} J(J+1) \text{ cm}^{-1} \qquad \text{(X)}$$

Where, c is the velocity of light, expressed in cms.

~~Eqn. (X) can be written as—~~

$$\varepsilon_J = BJ(J+1) \text{ cm}^{-1} \qquad \text{(XI)}$$

Where, B is called the rotational constant, which is given by

$$B = \frac{\hbar}{8\pi^2 I_B c} \qquad \text{(XII)}$$

In eqn. (XII), I_B is used instead of I, since for diatomic molecule, $I_A = 0$ & $I_B = I_C$. One can also use I_C, but the notation of eqn. (XII) is conventional.

TU-16 (C-16)

Q4. Show that, "the rotational energy gap between two levels increases with increasing the value of rotational quantum number."

⇨ The energy of a rotational level J is given by,

$$\varepsilon_J = BJ(J+1)\,cm^{-1} \longrightarrow (1)$$

Eqn. (1) states that, for $J=0$, $\varepsilon_0 = 0$, i.e. the molecule doesn't rotate at all. For $J=1$, $\varepsilon_1 = 2B\ cm^{-1}$ & the value of ε_J increases with increasing J value. But in practice there comes a point at which the centrifugal force of a rapidly rotating diatomic molecule is greater than the strength of the bond & the molecule is disrupted. This is illustrated in the following figure—

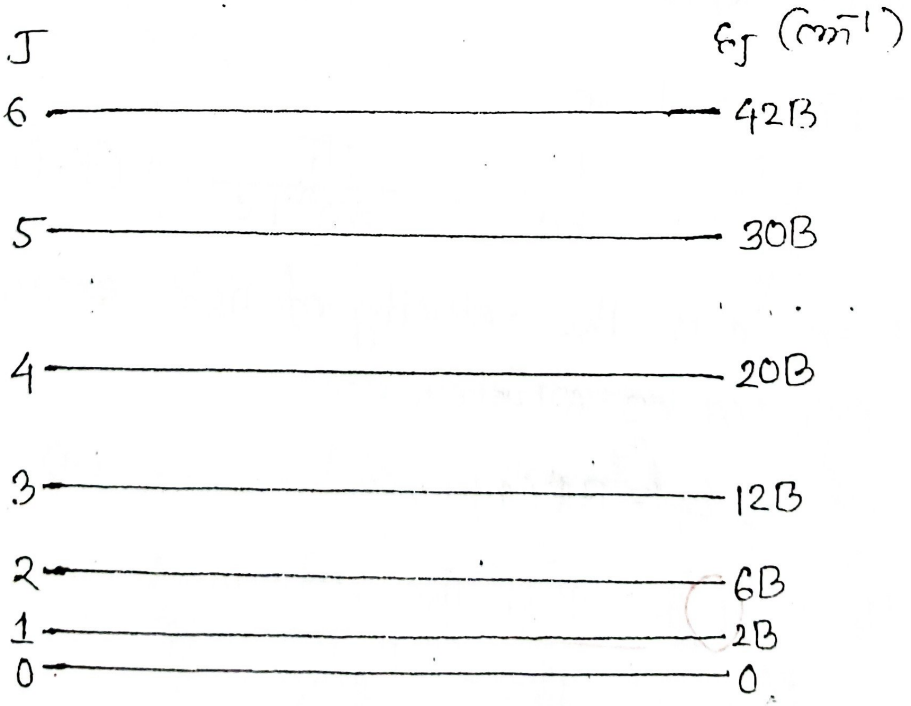


figure → The allowed rotational energy levels of a rigid diatomic molecule.

Let us consider transition of a molecule betn two distinct rotational energy levels. The energy change due to transition from $J=0$ state to $J=1$ state is-

$$\varepsilon_{J=1} - \varepsilon_{J=0} = 2B - 0 = 2B\ cm^{-1}.$$

& the corresponding wave number is —

$$\bar{\nu}_{J=0 \to J=1} = 2B\ cm^{-1}.$$

Similarly for transition betn $J=1$ state & $J=2$ state,

$$\bar{\nu}_{J=1 \to J=2} = \varepsilon_{J=2} - \varepsilon_{J=1}$$
$$= 6B - 2B$$
$$= 4B\ cm^{-1}.$$

In general to raise a molecule from J to $J+1$ we have,

$$\bar{\nu}_{J \to J+1} = \varepsilon_{J+1} - \varepsilon_J$$
$$\Rightarrow \bar{\nu}_{J \to J+1} = B(J+1)(J+2) - BJ(J+1) \quad ?$$
$$\Rightarrow \bar{\nu}_{J \to J+1} = 2B(J+1)\ cm^{-1} \quad \text{——————} \to (II)$$

That is the rotational energy gap (hence wave number gap) increases with increasing the value of J.

This phenomenon is illustrated in the following figure-

Let us consider transition of a molecule betn two distinct rotational energy levels. The energy change due to transition from $J=0$ state to $J=1$ state is-

$$\varepsilon_{J=1} - \varepsilon_{J=0} = 2B - 0 = 2B \text{ cm}^{-1}.$$

& the corresponding wave number is —

$$\bar{\nu}_{J=0 \to J=1} = 2B \text{ cm}^{-1}$$

Similarly for transition betn $J=1$ state & $J=2$ state,

$$\begin{aligned} \bar{\nu}_{J=1 \to J=2} &= \varepsilon_{J=2} - \varepsilon_{J=1} \\ &= 6B - 2B \\ &= 4B \text{ cm}^{-1}. \end{aligned}$$

In general to raise a molecule from J to $J+1$ we have,

$$\bar{\nu}_{J \to J+1} = \varepsilon_{J+1} - \varepsilon_J$$

$$\Rightarrow \bar{\nu}_{J \to J+1} = B(J+1)(J+2) - BJ(J+1)$$

$$\Rightarrow \bar{\nu}_{J \to J+1} = 2B(J+1) \text{ cm}^{-1} \quad \text{——————} \to (II)$$

That is the rotational energy gap (hence wave number gap) increases with increasing the value of J.

This phenomenon is illustrated in the following figure-

PTO.

04. Explain the selection rule. [Tu10, 2010,] Tu-13

⇨ **Selection rule:**

Those transitions are allowed in which the value of rotational quantum number, 'J' → changes by one unit, all other transitions are spectroscopically forbidden. Such a result is called a selection rule & we may (নিষিদ্ধ) formulate it as follows—

Selection rule: $\Delta J = \pm 1$

Tu-13. ~~Tu-16~~

05. Explain or write about 'the non-rigid rotator". Derive an expression for the spectrum of non-rigid rotator.

⇨ **Non-rigid rotator:**

All bonds are elastic to some extent. So, a molecule may have vibrational energy. i.e. the bond will stretch & compress periodically with a certain fundamental frequency dependent upon the masses of the atoms & the elasticity of the bond. If the motion is simple harmonic, then the force constant is given by—

$$K = 4\pi^2 \bar{\omega}^2 c^2 \mu$$

Where, $\bar{\omega}$ is the vibration frequency expressed in cm^{-1}. The vibration of B with J is determined by the force constant; the weaker the bond the more readily it will distort under centrifugal forces.

P.T.O.

Another consequence of elasticity is that the quantity r & B vary during a vibration. Many vibrations occur during a rotation & hence the measured value is the average. From the defining eqn. of B,

$$B = \frac{\hbar}{8\pi^2 Ic} = \frac{\hbar}{8\pi^2 c\mu r^2}$$

$$\Rightarrow B \propto \frac{1}{r^2}$$

The average value of $\frac{1}{r^2}$ is not equal to $\frac{1}{r_e^2}$, where, r_e is the equilibrium distance. In fact vibrations are not simple harmonic, since a real bond may be stretched more easily than it may be compressed & results $r_{av} > r_{eq}$.

(L) The spectrum of non rigid rotator: *

The Schrödinger wave eqn. may be set up for a non-rigid molecule & the rotational energy levels are found to be—

$$E_J = \frac{\hbar^2}{8\pi^2 I} J(J+1) - \frac{\hbar^4}{32\pi^4 I^2 r^2 K} J^2(J+1)^2 \text{ Joule} \quad \text{——— (1)}$$

$$\Rightarrow \bar{E}_J = \frac{E_J}{hc} = BJ(J+1) - DJ^2(J+1)^2 \text{ cm}^{-1} \quad \text{——— (1}$$

where, B is the rotational constant having the value of $\frac{\hbar}{8\pi^2 Ic}$ cm^{-1} & D is called the centrifugal distortion constant, which is given by—

P.T.O

$$D = \frac{h^3}{32\pi^4 I^2 r^2 k c} \text{ cm}^{-1} \qquad \text{---------- (III)}$$

Eqn. (II) is applied for a simple harmonic force field only; if the force field is an anharmonic, the expression becomes—

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + KJ^4(J+1)^4 + \cdots \text{ cm}^{-1}$$

$$\text{---------- (IV)}$$

Where, H & K etc are small constants dependent upon the geometry of the molecule. They are negligible compared with D ~~& most modern spectroscopic data are adequately fitted with eqn. (II)~~

From the defining eqns. of B & D, it may be shown that,

$$D = \frac{16B^3\pi^2\mu c^2}{K} = \frac{4B^3}{\bar{\omega}^2} \qquad \text{---------- (V)}$$

Where, $\bar{\omega}$ is the vibrational frequency of the bond & K is the force constant. For small J, the correction term $DJ^2(J+1)^2$ is almost negligible but for J values of 10 or more it may become appreciable. Following figure shows the comparable rotational energy levels of rigid & non-rigid rotator.

P.T.O.

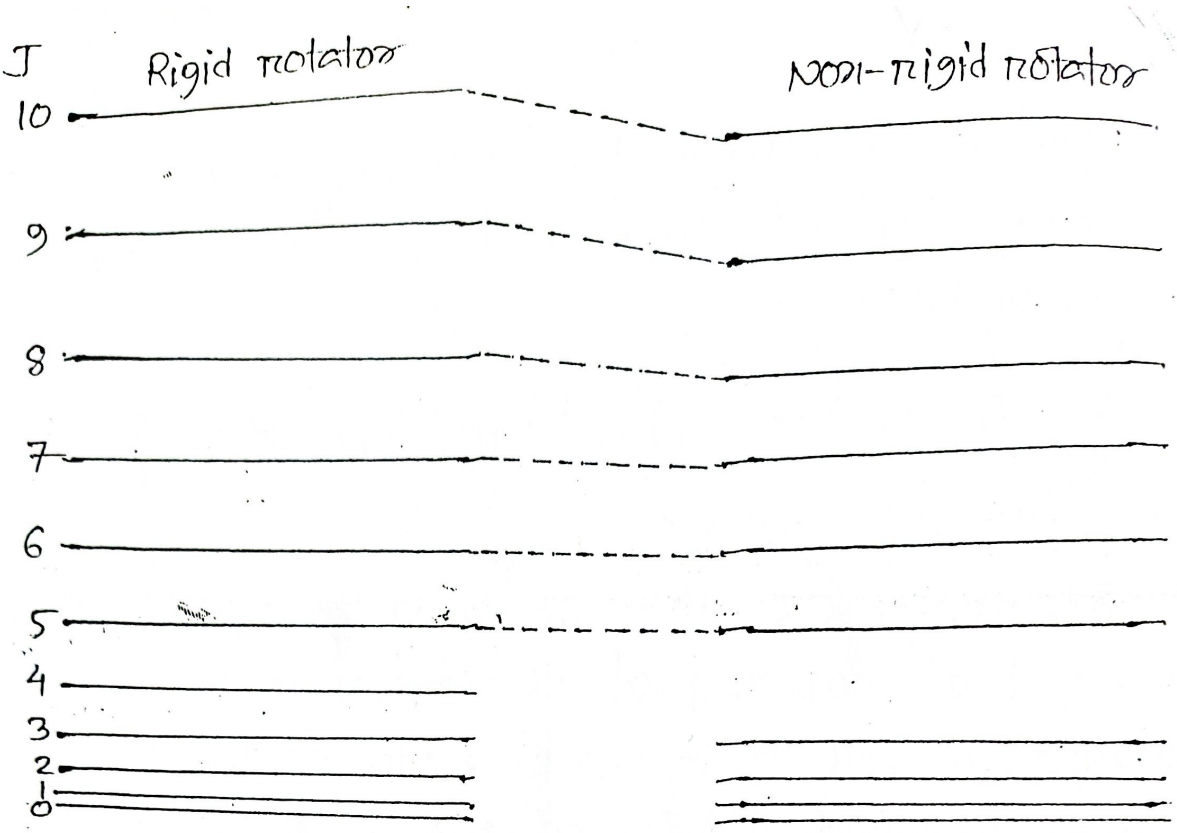


Figure → The comparable position of the rotational energy levels of rigid & non rigid diatomic molecule.

The comparable rotational spectra of rigid & non-rigid diatomic molecules are illustrated in the following figure-

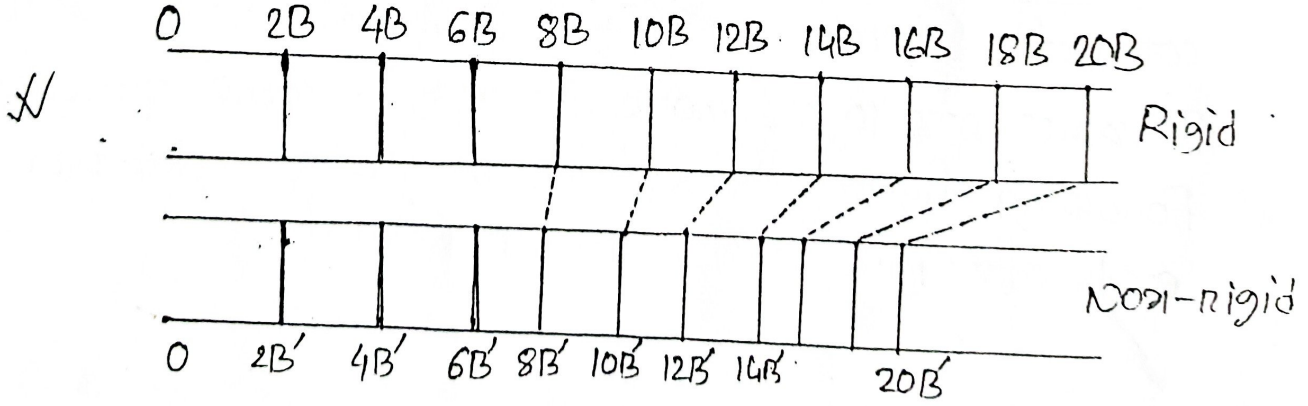


figure → comparable spectrum of rigid & non-rigid diatomic molecule.

P.T.O.

The analytical expression for the transition of a non-rigid molecule allowed by the selection rule is—

$$\varepsilon_{J+1} - \varepsilon_J = \bar{\nu}_J = B\left[(J+1)(J+2) - J(J+1)\right] - D\left[(J+1)^2(J+2)^2 - J^2(J+1)^2\right]$$

$$\Rightarrow \bar{\nu}_J = 2B(J+1) - 4D(J+1)^3 \text{ cm}^{-1} \quad \text{——(vi)}$$

The ~~knowledge of~~ D enables us to determine the vibrational frequency of a diatomic molecule as follows—

$$\bar{\omega}^2 = \frac{4B^3}{D}$$ & the force constant,

$$K = 4\pi^2 c^2 \bar{\omega}^2 \mu.$$

2013Q7. Compare the energy spacing betn the rigid & non-rigid rotator model. [2009, 06] [Tu-11]

➪ The energy expression for non-rigid rotator is—

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \text{ cm}^{-1}.$$

Now, transition from J to J+1

$$\bar{\nu}_J = \varepsilon_{J+1} - \varepsilon_J$$
$$= B\left[(J+1)(J+2) - J(J+1)\right] - D\left[(J+1)^2(J+2)^2 - J^2(J+1)^2\right]$$
$$= B(J+1).2 - D(J+1)^2\left[(J+2)^2 - J^2\right]$$
$$= 2B(J+1) - D(J+1)^2\left[J^2 + 4J - J^2 + 4\right]$$
$$= 2B(J+1) - 4D(J+1)^3.$$

P.T.O.

So, here we see that the energy spacing betn two successive energy levels is not exactly $2B\ cm^{-1}$, it is smaller than $2B\ cm^{-1}$.

Energy expression for rigid rotator is—

$$\varepsilon_J = BJ(J+1)\ cm^{-1}$$

Now, transition from J to $J+1$,

$$\begin{aligned}\bar{\nu}_J &= \varepsilon_{J+1} - \varepsilon_J \\ &= B[(J+1)(J+2) - J(J+1)] \\ &= B(J+1)(J+2-J) \\ &= 2B(J+1)\end{aligned}$$

So, here the energy spacing betn two successive energy level is $2B\ cm^{-1}$.

> figure: 2 (two) figure only from the previous questions & then the name of the two figure will be →

figure→ from (a) & (b) the change in energy level & rotational spectrum when passing from a rigid to a non rigid di-atomic molecule.

Q. Point out the three important characteristics /notifications of triatomic (or polyatomic) linear molecules.

⇨ Polyatomic linear molecules:

For polyatomic linear molecules $I_B = I_C$ & $I_A = 0$. So, the energy levels are given by,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + \dots\dots \text{ cm}^{-1} \quad \text{——(1)}$$

& the spectrum will show 2B separation modified by the distortion constant. The following three points for the polyatomic linear molecules should be noted —

- Since the moment of inertia for end-over-end rotation of a polyatomic linear molecule is considerably greater than that of a diatomic molecule, the B value will be much smaller & the spectral lines will be more closely spaced.

- The molecule must possess a dipole moment if it is to exhibit a rotational spectrum. Thus OCS (carbon oxysulphide) will be microwave active but $O^{16}C^{12}O^{18}$ is microwave inactive. Isotopic substitution doesn't lead to a dipole moment since the bond length & atomic charges are unaltered by the substitution. Thus $^{16}O^{12}C^{18}O$ is microwave inactive.

P.T.O.

A non-cyclic polyatomic molecule containing N number of atoms has N−1 individual bond length to be determined.

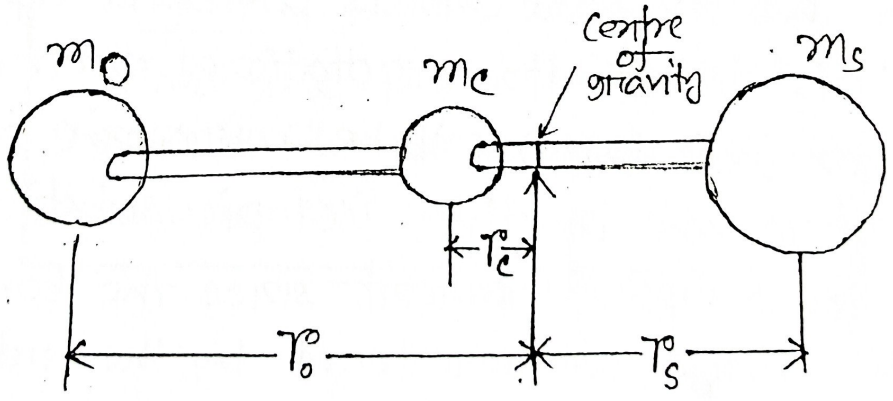
09. Derive an expression of rotational spectra for the polyatomic (or triatomic) linear molecules.

Or, Find out the bond length of C–O & C–S bonds in OCS molecule. [2010, 2009]

⇨ Let us consider the rotation of OCS molecule following figure shows the molecule where r_O, r_C & r_S represent the distances of the atoms from the centre of gravity. Consideration of moments gives,

$$m_O r_O + m_C r_C = m_S r_S \quad \longrightarrow (1)$$

Where, m_i is the mass of atom i.



Figure→ The molecule of OCS, showing the distances of each atom from the centre of gravity.

P.T.O.

The total moment of inertia is—

$$I = m_O r_O^2 + m_C r_C^2 + m_S r_S^2 \longrightarrow (II)$$

from the figure we have,

$$r_O = r_{CO} + r_C \quad \& \quad r_S = r_{CS} - r_C \longrightarrow (III)$$

Where, r_{CO} & r_{CS} represents the bond lengths of C–O & C–S bond respectively. Substituting in eqn (I) from eqn. (III) we have,

$$m_O (r_{CO} + r_C) + m_C r_C = m_S (r_{CS} - r_C)$$

$$\Rightarrow (m_O + m_C + m_S) r_C = m_S r_{CS} - m_O r_{CO}$$

$$\Rightarrow M r_C = m_S r_{CS} - m_O r_{CO} \longrightarrow (IV)$$

Where, M denotes the total mass of the molecule, Substituting eqn. (III) in eqn. (II), we have,

$$I = m_O (r_{CO} + r_C)^2 + m_C r_C^2 + m_S (r_{CS} - r_C)^2$$

$$\Rightarrow I = M r_C^2 + 2 r_C (m_O r_{CO} - m_S r_{CS}) + m_O r_{CO}^2 + m_S r_{CS}^2$$

$\longrightarrow (V)$

finally, substituting for r_C from eqn(IV) —

$$I = M \cdot \left(\frac{m_S r_{CS} - m_O r_{CO}}{M} \right)^2 + 2 r_C (m_O r_{CO} - m_S r_{CS}) + m_O r_{CO}^2 + m_S r_{CS}^2 .$$

P.T.O.

$$\Rightarrow I = \frac{(m_S r_{CS} - m_O r_{CO})^2}{M} + 2(m_O r_{CO} - m_S r_{CS}) \cdot \frac{m_S r_{CS}}{M}$$

$$+ m_O r_{CO}^2 + m_S r_{CS}^2$$

$$\Rightarrow I = \frac{(m_S r_{CS} - m_O r_{CO})^2}{M} - \frac{2(m_S r_{CS} - m_O r_{CO})^2}{M}$$

$$+ m_O r_{CO}^2 + m_S r_{CS}^2$$

$$\Rightarrow I = m_O r_{CO}^2 + m_S r_{CS}^2 - \frac{(m_S r_{CS} - m_O r_{CO})^2}{M}$$

$$\Rightarrow I = m_O r_{CO}^2 + m_S r_{CS}^2 - \frac{\{-(m_O r_{CO} - m_S r_{CS})\}^2}{M}$$

$$\therefore I = m_O r_{CO}^2 + m_S r_{CS}^2 - \frac{(m_O r_{CO} - m_S r_{CS})^2}{M} \quad \longrightarrow (VI)$$

Considering the isotopic molecule ^{18}OCS we can write m_O' for m_O in eqn (VI). Thus—

$$I' = m_O' r_{CO}^2 + m_S r_{CS}^2 - \frac{(m_O' r_{CO} - m_S r_{CS})^2}{M'} \quad \longrightarrow (VII)$$

Now, solving eqn (VI) & (VII), the values of r_{CO} & r_{CS} can be calculated.

10. Derive an expression of rotational spectra for the polyatomic symmetric top molecule.

P.T.O

⇨ Expression for the polyatomic symmetric top molecules:

In rotational spectral studies, a molecule is said to be symmetric top if it has two of its three principal moments of inertia equal & the third moment of inertia is non zero. i.e-

$$I_A \neq I_B = I_C \quad \& \quad I_A \neq 0.$$

We need two quantum numbers to describe the degree of rotation, one for I_A & one for I_B & I_C. However a quantum number J is considered to represent the total angular momentum of the molecule. It is conventional to use K to represent the angular momentum about the top axis.

The allowed values of both K & J are integral or zero. The total angular quantum number may have value from 0,1,2...∞ ; but the value of K never be larger than J. K may have negative values. For a given value of J & K may have following values—

$$K = J, J-1, J-2, \ldots, 0, \ldots, -(J-2), -(J-1), -J \qquad \longrightarrow (1)$$

Which is a total of $(2J+1)$ values.

If we take first the case of rigid symmetric top then the Schrödinger eqn. may be solved to give

P.T.O.

the allowed energy levels for rotation as—

$$\varepsilon_{J,K} = \frac{E_{J,K}}{hc} = BJ(J+1) + (A-B)K^2 \ cm^{-1}$$ ——— (II)

Where, $A = \frac{h}{8\pi^2 I_A c}$ & $B = \frac{h}{8\pi^2 I_B c}$

The selection rules for this molecule—

$$\Delta J = \pm 1 \ \& \ \Delta K = 0$$ ——— (III)

for transition betn J & J+1 we have,

$$\varepsilon_{J+1} - \varepsilon_{J,K} = \bar{\nu}_{J,K} = B(J+1)(J+2) + (A-B)K^2 - [BJ(J+1) + (A-B)K^2]$$

$$\Rightarrow \bar{\nu}_{J,K} = 2B(J+1) \ cm^{-1}.$$ ——— (IV)

Thus, the spectrum is independent of K, & hence rotational changes about the symmetry axis do not give rise to a rotational spectrum.

11. Explain the term— "Microwave-oven".

⇨ Microwave oven:

The mode of operation of microwave oven entirely depends upon the absorption of microwave radiation by food. In fact water molecules only absorbs microwave radiation & become raised to

PTC

higher rotational states. The biological molecules in food are too large to rotate. As with many other excited states, the excess rotational energy of the water molecules is re-emitted as heat & the food becomes cooked.

In microwave oven, water molecules throughout the whole bulk of the food are simultaneously excited, so, cooking time is drastically reduced.

12. Explain about the "techniques & instrumentation" for a microwave spectrometer.

Or, Draw schematic diagram of a microwave spectrometer by explaining its different part. [Tu 07] 4

⇨ Techniques & Instrumentation:

The following figure shows how various components of a microwave spectrometer are linked together →

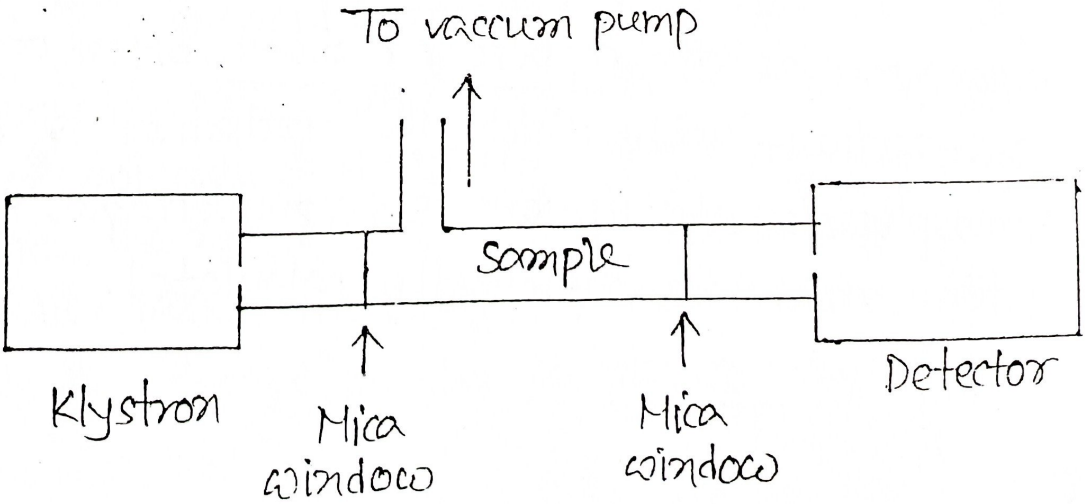


Figure → Schematic diagram of a microwave spectrometer

(I) The source & monochromator:

(The usual source of this region is the klystron valve which emits radiation over only a very narrow frequency range.) Since it can act as a monochromatic source. The actual emission frequency is variable & so a spectrum may be scanned over a limited range of frequencies using a single klystron.)

One slight disadvantage of this source is that the total energy radiated is very small. Since, this is all concentrated into a very narrow range of frequencies, the electric field generated at this frequencies is more intense than might be expected from the radiative power. It means - a sharply tuned detector can be sufficiently sensitive to detect the signal.

(II) Beam direction:

This is achieved by the use of "waveguides", hollow tubes of copper or silver, often of rectangular cross-section inside which the radiation is confined. Atmospheric absorption is considerable, so the whole system must be efficiently calculated.

P.T.O.

(III) Sample space:

Normally the sample is gaseous & is retained in a piece of evacuated wave guide by very thin mica windows. Very small pressures are sufficient to obtain a spectrum & many substances which are normally thought of as liquid or solid have sufficient vapour pressure to allow them to be studied by the technique.

(IV) Detector:

It is possible to use a radio receiver as detector. A simple crystal detector is found to be more sensitive & easier to use.

To-15

13. Discuss the effect of isotopic substitution on rotational spectra. [2009, 2007, 2006,] (5)

P-14

P-14 ⇨ The effect of isotopic substitution:

The replacement of an atom by its isotope doesn't affect the bond length & chemical properties of the molecule but result the change in total mass & hence the moment of inertia & B value for the molecule. Therefore change of an atom by its isotope results the change in the position of the signal.

P.T.O.

For example, if we consider the case of carbon monoxide molecule. On going from $^{12}C^{16}O$ to $^{13}C^{16}O$ there is a mass increase & hence a decrease in the B value. If we mention the ^{13}C molecule with a prime we have $B > B'$.

Where, $B = \frac{\hbar}{8\pi^2 Ic}$ & $B' = \frac{\hbar}{8\pi^2 I'c}$; $\boxed{I = mr^2}$

This change reflects in the rotational energy levels of the molecule. The following figure shows the relative lowering of the rotational energy levels for ^{13}C with respect to those for ^{12}C.

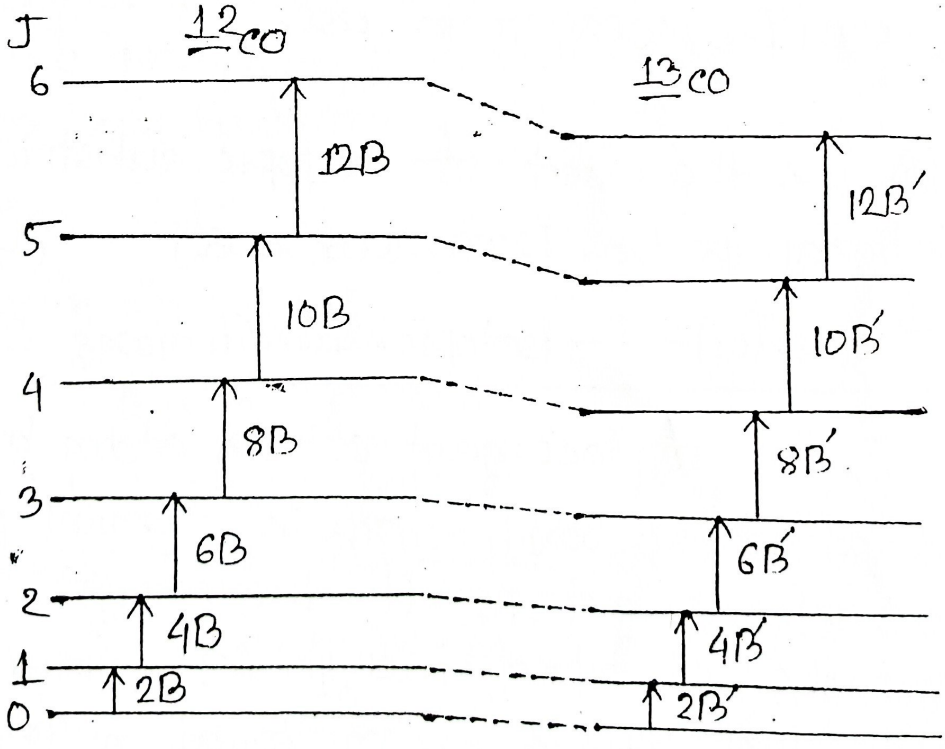


figure → The effect of isotopic substitution on the energy level.

P.T.O.

The following figure shows the comparable spectrum of ^{12}CO & ^{13}CO. The figure shows that, the spectrum of heavier species will show a smaller separation betn the lines (2B') than that of the higher one (2B).

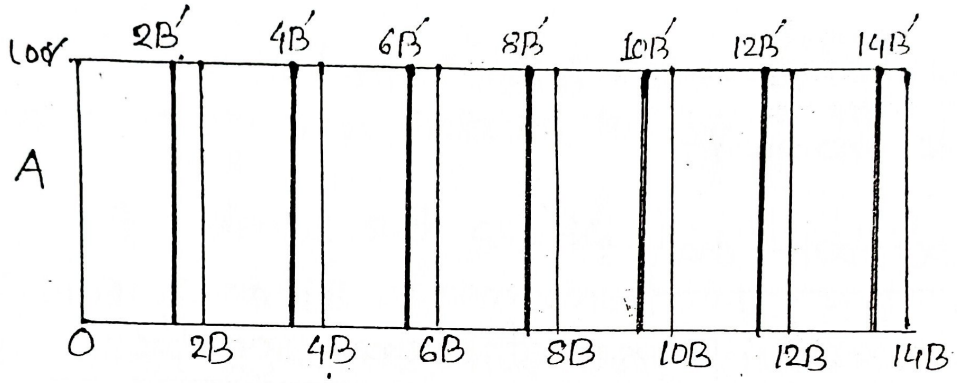


figure: Comparable spectrum of ^{12}CO & ^{13}CO.
The Green lines represent the spectral lines of ^{13}CO.

Various Important Notes from the class lecture

☐ Why $B_{polyatomic} < B_{Diatomic}$?

⇨ Moment of inertia যদি ছোট হয়, তাহলে B বড় হবে। আমরা জানি, Polyatomic molecule এর ক্ষেত্রে moment of inertia বড়, তাই B ছোট। So, $B_{polyatomic} < B_{diatomic}$.

আবার, $\underset{\text{(Polyatomic molecule)}}{I_B = I_C} > \underset{\text{(Diatomic molecule)}}{I_B = I_C}$; Here, I = moment of inertia

- OCS ⟶ Microwave active

  $^{16}O C^{18}O$ ⟶ Microwave inactive

  $^{18}O C^{16}O$ ⟶ Microwave inactive.

- Microwave spectrum থেকে bond length নির্ণয় করা যায়।

- Isotopic substance doesn't lead to change of dipole moment.

- Important chart:

Microwave spectrum of carbon oxysulfide:

$J \rightarrow J+1$	$\bar{\nu}_{obs.}$ (cm^{-1})	$\Delta\bar{\nu}$	B (cm^{-1})
0 → 1	.....		
		2X0.4055	0.2027
1 → 2	0.8109		
		0.4054	0.2027
2 → 3	1.2163		
		0.4054	0.2027
3 → 4	1.6217		
		0.4054	0.2027
4 → 5	0.0271		

—— D ——

Joy
CHEM-38

Q1. (AS) What is the basic requirement of a molecule to absorb microwave radiation? Explain the interaction of radiation with rotating molecule. [Tu10, 2009, 2006] [Tu-11] [Tu-14, 15]

⇨ The permanent electric dipole moment is the basic requirement of a molecule to absorb microwave radiation.

A molecule such as HCl in which one atom (the hydrogen) carries a permanent net positive charge & the other a net negative charge, is said to have permanent electric dipole moment. If we consider the rotation of HCl with the interaction of radiation. then—

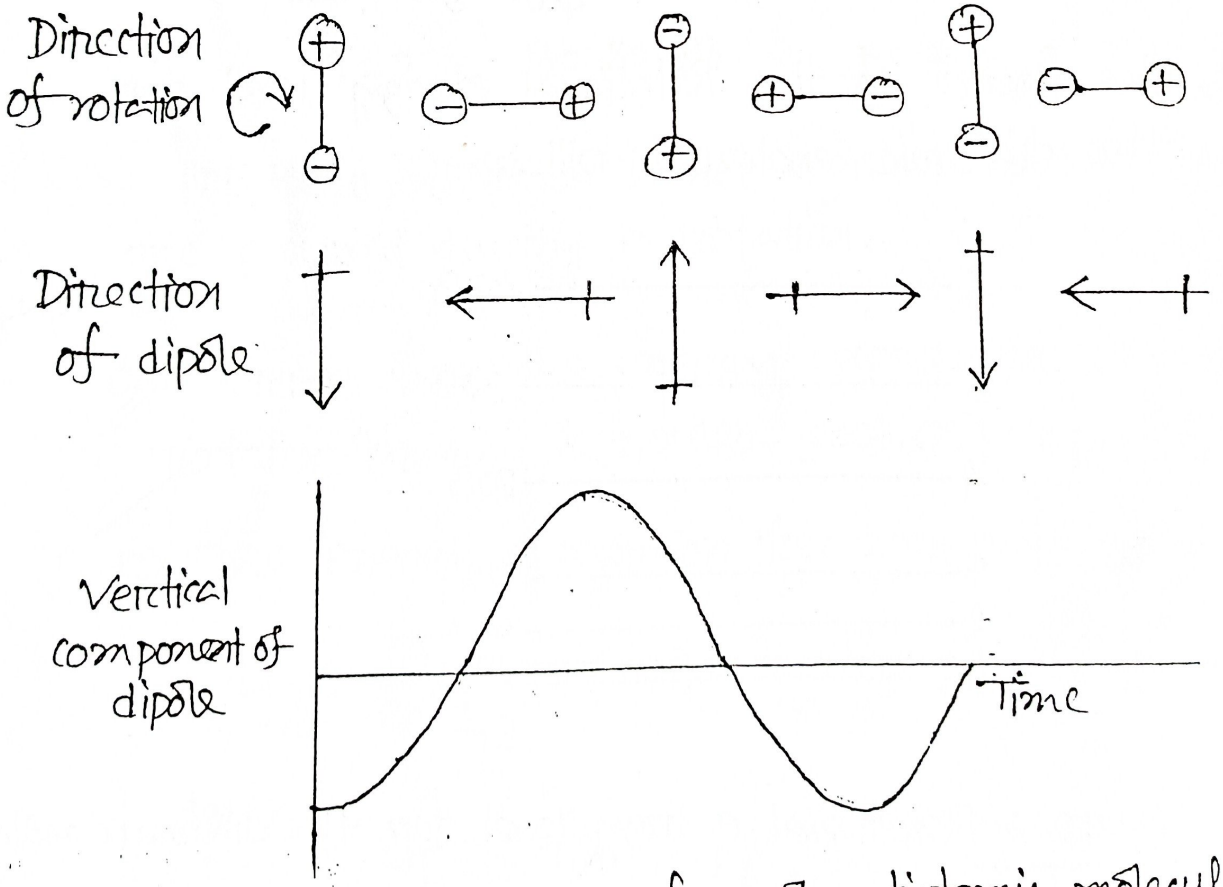


figure → The rotation of a polar diatomic molecule, showing the fluctuation in the dipole moment measured in a particular direction.

Q2. Draw the allowed rotational energies level of a rigid diatomic molecule.

➪ We know the energy expression for rigid diatomic molecule in cm^{-1} is—

$$\epsilon_J = BJ(J+1) \longrightarrow (1)$$

For, $J=0$; $\epsilon_J = 0$.

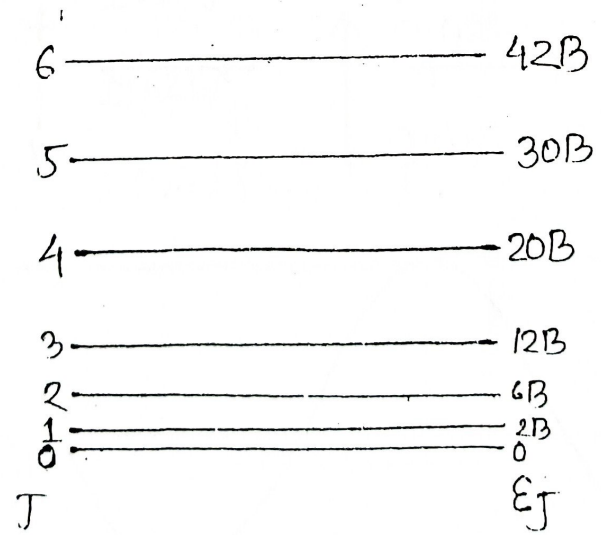
We can say molecule is not rotating at all.

For, $J=1$; Rotational energy, $\epsilon_J = B.1(1+1)$
$= 2B$.

For, $J=2$; Rotational energy, $\epsilon_J = 6B$.

For, $J=3$; Rotational energy, $\epsilon_J = 12B$.

So, the figure of the Rotational energy level for the diatomic molecule will be—



figure→ Rotational energy level for the diatomic molecule

03. What are the limitations of rigid rotator model. [2009, 06] (AS)

➪ The limitations of rigid rotator model:

(I) The bond betn two atoms is considered as rigid bond in rigid rotator model which is not correct since all bonds are elastic to some extent.

(II) It is considered that the spectrum betn successive two lines are 2B cm^{-1} which is not correct. The seperation betn successive lines decrease steadily with increasing I.

For H-F,

$J=0$; $B=20.56$.

$J=1$; $B=20.48$.

$J=2$; $B=20.43$.

(III) Here vibration is not considered, Many vibrations are occured during a rotation.

04. Write down the energy expression for non-rigid rotator in cm^{-1} & joules. [2009, 06] (AS)

➪ The energy expression for non-rigid rotator is—

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) - \frac{h^4}{32\pi^4 I^2 r_0^2 K} J^2 (J+1)^2 \text{ joules.}$$

Again, $$\epsilon_J = \frac{E_1}{hc}$$

$$= BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1}.$$

Where, B = Rotational constant ; D = centrifugal distortion constant.

05. What is the centrifugal distortion constant? Write down the significance of it in non-rigid rotato.

(As)

⇨ The centrifugal distortion constant is denoted by D; which is given by—

$$D = \frac{\hbar^3}{32\pi^4 I^2 \bar{\nu}^2 kc} \text{ cm}^{-1}.$$

which is a positive quantity.

The significance are given below—

(i) It allocas us to determine the value of lines in an observed spectrum.

(ii) D enable to determine the vibrational frequency of diatomic molecule.

$$D = \frac{4B^3}{\bar{\omega}^2}$$

Where,

B = Rotational constant

$\bar{\omega}$ = The vibrational frequency of the bond.

06. What is the rotational constant? [2009, 06] [Tu-11]

⇨ The rotational constant, "B" is given by—

$$B = \frac{\hbar}{8\pi^2 I_B c}$$

in which we have used explicitly the moment of inertia, I_B.

07. Write down the energy equation for a rigid symmetric top molecule. Explain using the equation why for $K>0$, the rotational energy levels are degenerate. [Tu 10, 2010, 2005] (AS)

⇨ Energy expression for a rigid symmetric top molecule in cm^{-1} is—

$$\varepsilon_{J,K} = \frac{E_{J,K}}{hc}$$

$$= BJ(J+1) + (A-B)K^2 \ cm^{-1}.$$

Where, J = Total angular momentum

K = Angular momentum about the top axis.

$K = J, J-1, J-2 \ldots\ldots 0 \ldots\ldots -(J-1), -J$

$$B = \frac{h}{8\pi^2 I_B c} \quad \& \quad A = \frac{h}{8\pi^2 I_A c}$$

We see that, energy depends on K^2. So that it is immaterial whether the top spins clockwise or anticlockwise; the energy is the same for a given angular momentum. For all $K>0$, Therefore the rotational energy levels are doubly degenerate.

08. What is the energy expression for the rotational transition in rigid symmetric top molecule. How is it changed for non-rigid molecule. [2005]

P.T.O.

⇨ The energy expression for the rotational tra of a rigid symmetric top molecule,

$$\varepsilon_{J,K} = \frac{E_{J,K}}{hc}$$

$$= BJ(J+1) + (A-B)K^2 \ cm^{-1}$$

When, centrifugal stretching is taken into account, the energy levels becomes —

$$\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2 - D_J J^2 (J+1)^2 - D_{JK} J(J+1)K^2 - D_K K^4 \ cm^{-1}.$$

Which is the energy expression for non-rigid top symmetric molecule.

2012 2009. Show the absorption spectrum of a rigid rotator is expected to consist of a series of a equidistance lines. [2006] (AS)

⇨ The energy expression for rigid diatomic molecule —

$$\varepsilon_J = BJ(J+1) \ cm^{-1} \quad \text{——(1)}$$

If we want to rise the molecule from $J=0$ to $J=1$ then the energy absorbed

$$\varepsilon_{J=1} - \varepsilon_{J=0} = B.1(1+1) - B.0(0+1)$$

$$= 2B \ cm^{-1}.$$

If the rises of the molecule from $J=1$ to $J=2$ then,

P.T.O.

$$\varepsilon_{J=2} - \varepsilon_{J=1} = B.2(2+1) - B.1(1+1)$$
$$= 6B - 2B$$
$$= 4B \text{ cm}^{-1}.$$

In general to rise the molecule from J to J+1,

$$\bar{\nu}_{J \to J+1} = \varepsilon_{J+1} - \varepsilon_J$$
$$= B(J+1)(J+2) - BJ(J+1)$$
$$= 2B(J+1) \text{ cm}^{-1}.$$

Therefore,
when molecule rise from J=0 to J=1 then,

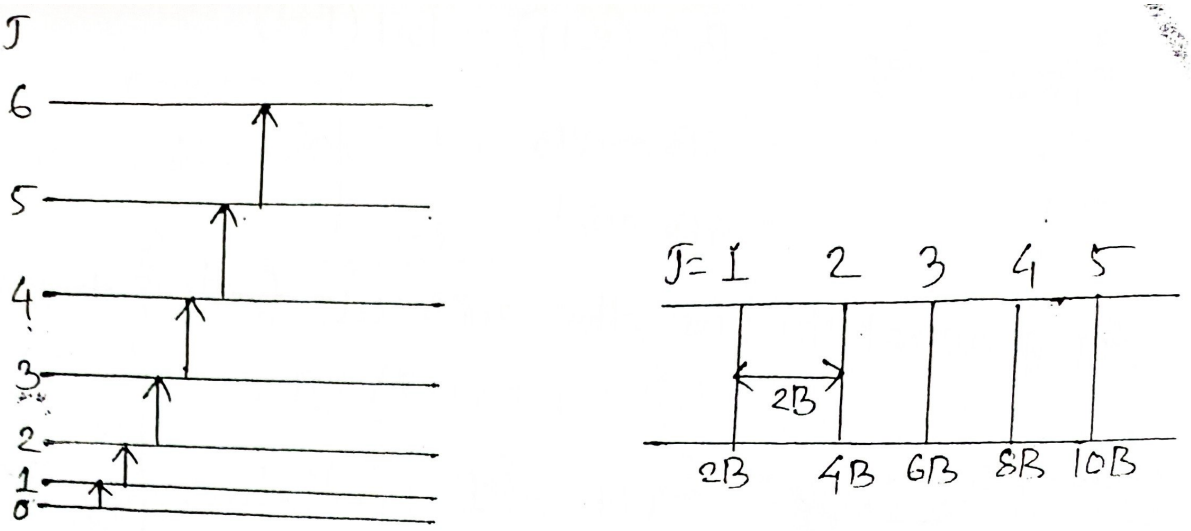
$$\bar{\nu}_{J=0 \to J=1} = 2B(1+1) - 2B(0+1)$$
$$= 4B - 2B$$
$$= 2B$$

That means, we will get absorption spectrum at $2B \text{ cm}^{-1}$. Similarly,

$$\bar{\nu}_{J=1 \to J=2} = 2B(2+1) - 2B(1+1)$$
$$= 6B - 4B$$
$$= 2B$$

$$\therefore \bar{\nu}_{J=2 \to J=3} = 2B(3+1) - 2B(2+1)$$
$$= 8B - 6B$$
$$= 2B.$$

P.T.O.



figure→ Allowed transitions betn the energy levels of a rigid diatomic molecule & the spectrum which arises for them.

10. classify the molecule according to the principal moment of inertia. [TU 07, 2004]

⇨

C_6H_6 → spherical top	SF_6 → spherical top.
BCl_3 → symmetric top.	CH_3F → Symmetric top.
CH_3Cl → Symmetric top	HCN → linear
NH_3 → Symmetric top	CO_2 → linear
H_2O → Asymmetric top	CH_4 → spherical top
C_2H_4 → Asymmetric top	CH_2Cl_2 → Asymmetric top.
C_2H_2 → linear.	

11. Write down the energy expression for the rigid gaseous diatomic molecule. How it is modified when the molecule is non rigid? [2009, TU 07, 05,]

⇨

P.T.O.

⇨ Energy expression for the rigid gaseous diatomic molecule in Joule is ⟶

$$E_J = \frac{\hbar^2}{8\pi^2 I} J(J+1) \text{ Joule.}$$

In terms of wave no. ⟶

$$\varepsilon_J = \frac{E_J}{hc} = \frac{\hbar}{8\pi^2 I c} J(J+1)$$

$$= BJ(J+1) \text{ cm}^{-1}.$$

[where, B = rot. constant $= \frac{\hbar}{8\pi^2 I_B c}$

$J = 0, 1, 2 \ldots$

In case of non-rigid rotator, a term centrifugal distortion constant (D) included, which is not negligible for non-rigid molecule & must be substract. Thus modified ⟶

So, energy expression for non-rigid molecule in cm^{-1}

$$\varepsilon_J = \frac{E_J}{hc} = BJ(J+1) - DJ^2(J+1)^2 \text{ cm}^{-1}$$

Where, B = rotational constant.

& D = centrifugal distortion constant

$$= \frac{\hbar^3}{32\pi^4 I^2 \hbar^2 K c}.$$

12. Assuming the molecule to be rigid, calculate the energy of the levels for $J=3$ & $J=4$. [2007]

⇨ We know, the energy expression for rigid molecule in cm^{-1} ; $\varepsilon_J = BJ(J+1)$

P.T.O.

So, when, $J=3$; then, $E_J = B.3(3+1) = 12B$.

When, $J=4$; then, $E_J = B.4(4+1) = 20B$.

13. Why N_2 & O_2 do not give rotational spectrum, or microwave spectrum, while HCl & CO give?

⇨ N_2 & O_2 are homonuclear molecule. So, during rotation, there is no change in dipole & no interaction with radiation hence not give rotational spectrum.

On the otherhand, HCl & CO are asymmetric or heteronuclear molecule, therefore dipole moment will change during rotation & hence give rotational or microwave spectrum.

14. Why the rotation about the bond axis is rejected?

⇨ There are two reasons for which the rotation about the bond axis is rejected.

(1) The moment of inertia about the bond is very small. We know,

$$E_J = \frac{\hbar^2}{8\pi^2 I} J(J+1) \text{ Joules} \quad (J=0,1,2\ldots)$$

or, $$E_J = \frac{\hbar}{8\pi^2 Ic} J(J+1) \text{ cm}^{-1} \quad (J=0,1,2\ldots)$$

from this eqn. we see that energy levels would be extremely widely spaced. So, a molecule requires a great deal of energy to be raised from $J=0$ to $J=1$ state &

P.T.O.

such transitions do not occur under normal spectroscopic condition. Thus diatomic molecule in J=0 state said to be not rotating.

(ii) There will be no dipole change, even if such a transition occurs & hence no spectrum.

15. Why the real vibrations are not simple harmonic?

⇨ The real vibrations are not simple harmonic because a real bond may be stretched more easily than it may be compressed & this usually results in r_{avg} being greater than r_{eq}.

16. Which of the following molecules will show a microwave rotational spectrum: [2010,Tu10]

H_2, CH_4, HCl, CH_3Cl, CH_2Cl_2, H_2O, SF_6

⇨ HCl, CH_3Cl, CH_2Cl_2, H_2O will show microwave rotational spectrum. [For explanation, see the additional ques. no-01]

Note & Hints:

Microwave spectrum দিতে হলে permanent dipole moment থাকতে হবে। অর্থাৎ permanent net positive charge & permanent net negative charge থাকতে হবে।

17. Write an expression ($\bar{\nu}_{J,K}$) for the spectrum of the molecule. Show the spectrum is independent of K from the expression. [Tu10.]

⇨ The expression ($\bar{\nu}_{J,K}$) for the rotational spectrum of the symmetric top molecule will be—

$$\bar{\nu}_{J,K} = B(J+1)(J+2) + (A-B)K^2 - [BJ(J+1) + (A-B)K^2]$$

When, we take $J \neq 0$ then, (by the selection rule)

$$\bar{\nu}_{J,K} = 2B(J+1) \text{ cm}^{-1}.$$

Thus, the spectrum is independent of K, & hence rotational changes about the symmetry axis do not give rise to a rotational spectrum.

Not needed

The reason for this, is quite evident— rotation about the symmetry axis does not change the dipole moment perpendicular to the axis (which always remains zero), & hence the rotation can't interact with radiation.

18. Write short notes on— Rotational spectra & bond length of diatomic molecule. [2009]

⇨ Write from the class lecture ques. no- 3 & 8.

19. Write short notes on – (i) source, (ii) wave guides, (iii) sample (iv) Detector in microwave spectroscopy. [Tu07]

⇨ See the class lecture ques. No 11.

20. What are rigid molecules? Why does a molecule become non-rigid during rotation at higher energies? [2008]

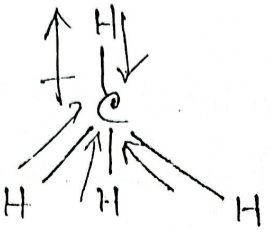
⇨ ✓

21. Explain the reason why CH_4 does not absorb MW radiation but absorb IR radiation. [2008]

⇨ All molecules having a permanent dipole moment are said to be 'microwave active'. CH_4 does not contain any permanent net negative charge. As why it doesn't contain any permanent electric dipole moment. so it does not absorb any MW radiation.

P.T.O.

But, there is a change of dipole moment for the CH_4 molecule. We know, the molecules which show the change of dipole moment are said to be IR active.



$\mu \neq 0$.

As why, CH_4 absorbs IR radiation.

22. Explain the reason why the rotational constants (B) of ^{12}CO & ^{13}CO are $1.92\ cm^{-1}$ & $1.84\ cm^{-1}$ respectively (100%) [2008]

⇨ If we consider the two isotopes of carbon (^{12}C & ^{13}C), then, for the increased mass of the isotope, the moment of inertia will be increased. All for the above reasons, the value of the rotational constant will be decreased.

So, the reason why the rotational constants (B) of ^{12}CO & ^{13}CO are $1.92\ cm^{-1}$ & $1.84\ cm^{-1}$ is nothing but isotopic (mass) effect.

Note: see the important notes no. (1) from the class lecture. >> mass কমা হলে (isotopic), moment of inertia কমে যায়, ফলে rotational constant (B) value বেড়ে যায়।

Joy
CHEM-3S

Mathmatical problems & solutions

Previous Years Questions with solutions:

(2010) 01. The following table shows some of the lines in the M.W. spectrum of $H^{19}F$ at a certain temperature.

$\bar{\nu}_{obs}(cm^{-1})$	41.08	82.19	123.15	164.00	204.62	244.93

(a) calculate the spacing betn the lines & calculate the rotational constant values from these. Explain, why the values are different.

(b) Assuming an average spacing of $41.1\ cm^{-1}$ betn the lines. Calculate the bond length of the molecule ($\mu_{H^{19}F} = 1.578 \times 10^{-27}\ kg$). [Tu10, 2010, 2007]

⇨ (a)

Obs.no.	$\bar{\nu}_{obs}(cm^{-1})$	$\Delta\bar{\nu}_{obs}(cm^{-1})$	$B(=\frac{1}{2}\Delta\bar{\nu}_{obs})$
1.	41.08		
		(82.19 − 41.08) = 41.11	20.56
2.	82.19		
		(123.15 − 82.19) = 40.96	20.48
3.	123.15		
		(164.00 − 123.15) = 40.85	20.43
4.	164.00		
		(204.62 − 164.00) = 40.62	20.31
5.	204.62		
		(244.93 − 204.62) = 40.31	20.16
6.	244.93		

P.T.O.

The variation of rotational constant (B) value can be explained as, the separation betn successive lines (& hence apparent B value) decreases steadily with increasing J. So, as one go upper level .i.e. increasing rotational quantum no. J, B values decreases.

(b) Here, $2B = 41.1\ cm^{-1}$

$\therefore B = 20.56\ cm^{-1}$.

$\mu_{H^{19}F} = 1.578 \times 10^{-27}\ kg$

We know,

$$I = \frac{\hbar}{8\pi^2 B.c.} = \frac{6.626 \times 10^{-34}}{8\pi^2 \times 20.56 \times 3 \times 10^{10}}$$

$$= 1.36 \times 10^{-47}\ kg\ m^2.$$

So, $r^2 = \frac{I}{\mu_{H^{19}F}}$

$$= \frac{1.36 \times 10^{-47}}{1.578 \times 10^{-27}}$$

$$= 8.62 \times 10^{-21}$$

$\therefore r = 9.28 \times 10^{-11}\ m.$ (Ans)

02. The $J=0 \rightarrow J=1$ rotational absorption line occurs at 1.153×10^{11} cycles per sec in $^{12}C^{16}O$ & at 1.102×10^{11} cycles per sec. in $^{x}C^{16}O$. Calculate the mass no 'x' of the unknown carbon isotope ^{x}C. [2009] [2006]

P.T.O

⇨ Here,

$$\frac{B}{B'} = \frac{1.153\times10^{11}}{1.102\times10^{11}} = \frac{\hbar}{8\pi^2 I c}\cdot\frac{8\pi^2 I' c}{\hbar} = \frac{I'}{I} = \frac{\mu'}{\mu}$$

$$= 1.046.$$

$$\therefore \frac{\mu'}{\mu} = 1.046$$

$$\Rightarrow \frac{m_1 m_2'}{m_1 + m_2'} \times \frac{m_1 + m_2}{m_1 m_2} = 1.046.$$

$$\Rightarrow \frac{16\, m_2'}{16 + m_2'} \times \frac{16+12}{16\times12} = 1.046$$

$$\Rightarrow \frac{16\, m_2'}{16 + m_2'} = 7.17$$

$$\Rightarrow 16 m_2' - 7.17 m_2' = 114.72$$

$$\Rightarrow 8.83 m_2' = 114.72$$

$$\Rightarrow m_2' = \frac{114.72}{8.83}$$

$$\therefore m_2' \simeq 13.$$

So, $X = 13$.

& then, $X_C{}^{16}O = {}^{13}C{}^{16}O$. (Ans)

03. The diatomic molecule HCl has B value of 10.593 cm^{-1} & centrifugal distortion constant D of $5.3\times10^{-4}\ cm^{-1}$. Estimate the vibrational frequency & force constant of the molecule.

[TU07]

P.T.O.

➪ We know,

$$\bar{\omega}^2 = \frac{4B^3}{D} = \frac{4\times(10.593)^3}{5.3\times10^{-4}}$$

$$= 8.97\times10^6\ cm^{-2}$$

$$\therefore \bar{\omega} = 2.995\times10^3\ cm^{-1}$$

Here,
$B = 10.593\ cm^{-1}$
$D = 5.3\times10^{-4}\ cm^{-1}$
$\bar{\omega} = ?$
$K = ?$

Now, reduced mass,

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$= \frac{1.673\times10^{-27}\times58.93\times10^{-27}}{(1.673+58.93)\times10^{-27}}$$

$$= 1.63\times10^{-27}\ kg$$

Again,
for H, $(m_1) = 1.008\ amu$
$= 1.008\times1.66\times10^{-27}\ kg$
$= 1.673\times10^{-27}\ kg.$

for, Cl, $(m_2) = 35.5\ amu$
$= 35.5\times1.66\times10^{-27}$
$= 58.93\times10^{-27}\ kg$

So, force constant,

$$K = 4\pi^2\bar{\omega}^2c^2\mu$$

$$= 4\times\pi^2\times(2.995\times10^3)^2\times(3\times10^{10})^2\times1.63\times10^{-27}$$

$$= 519.5\ kgs^{-2}$$

$$= 519.5\ Nm^{-1}.$$ (Ans)

04. The spacing of lines in the microwave spectra of HI is $12.8\ cm^{-1}$. Calculate the moment of inertia of the molecule. [Tu07]

P.T.O.

⇨ We know,

$$B = \frac{h}{8\pi^2 I c}$$

$$\therefore I = \frac{h}{8\pi^2 B c}$$

$$= \frac{6.626 \times 10^{-34}}{8 \times \pi^2 \times 6.4 \times 3 \times 10^{10}}$$

$$= 4.37 \times 10^{-47} \text{ kgm}^2$$

(Ans)

Here,

$2B = 12.8 \text{ cm}^{-1}$

$\therefore B = 6.4 \text{ cm}^{-1}$

Planck's constant,

$h = 6.626 \times 10^{-34}$ J.s.

$c = 3 \times 10^{10} \text{ cms}^{-1}$

Q5. Following data is available for HF rotational spectrum.

J	$\bar{\nu}_{obs}$ (cm^{-1})	$\Delta\bar{\nu}_{obs}$	$B = \frac{1}{2}\Delta\bar{\nu}_{obs}$	$r = \sqrt{\frac{I}{\mu}}$ (nm)
0	41.08			
1	82.19			
2	123.15			
3	164.00			
4	204.62			
5	244.93			

Calculate-

(a) The separation between the lines [Tu09]

(b) The value of B in each case.

(c) The reduced mass of HF.

(d) The bond length of HF.

(e) The intensity & the value of B were not the same each case. Explain, why? → $B_{12} = B\alpha\frac{1}{T}$

PT.O

⇨

(a) & (b) ↴

J	$\bar{\nu}_{obs}$ (cm^{-1})	$\Delta\bar{\nu}$	$B(=\frac{1}{2}\Delta\bar{\nu}_{obs})$	$r(=\sqrt{\frac{I}{\mu}})$ (nm)
0	41.08			
1	82.19	(82.19 − 41.08) = 41.11	20.56	9.28×10^{-2}
2	123.15	40.96	20.48	9.30×10^{-2}
3	164.00	40.85	20.43	9.31×10^{-2}
4	204.62	40.62	20.31	9.33×10^{-2}
5	244.93	40.31	20.16	9.37×10^{-2}

(c) Reduced mass for HF.

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{1.673\times10^{-27}\times31.54\times10^{-27}}{1.673\times10^{-27}+31.54\times10^{-27}}$$

$$= 1.58\times10^{-27}\ kg$$

For (H) → $m_1 = 1.008$ amu
$= 1.008\times1.66\times10^{-27}$ kg
$= 1.673\times10^{-27}$ kg

For, m_2(F) = 19 amu
$= 19\times1.66\times10^{-27}$ kg
$= 31.54\times10^{-27}$ kg

(d) Bond length,

$$r = \sqrt{\frac{I}{\mu}}$$

$$= \sqrt{\frac{1.36\times10^{-47}}{1.58\times10^{-27}}}$$

$$= 9.28\times10^{-11}\ m$$

$$= 9.28\times10^{-11}\times10^{9}\ nm$$

$$= 9.28\times10^{-2}\ nm.$$

$$I = \frac{h}{8\pi^2 Bc}$$

$$= \frac{6.626\times10^{-34}}{8\pi^2\times20.56\times3\times10^{10}}$$

$$= 1.36\times10^{-47}\ kg m^2$$

(e) see math prob no (1).

06. The frequency separation of the first two lines in the rotational spectrum of CO is $3.84\ cm^{-1}$. Calculate the bond length of CO ($\mu = 11.38 \times 10^{-27}\ kg$) [2008]

➪ Here,

$2B = 3.84\ cm^{-1}$

$\therefore B = 1.92\ cm^{-1}$

$\mu = 11.38 \times 10^{-27}\ kg$

$h = 6.626 \times 10^{-34}\ Js$

$c = 3 \times 10^{10}\ cms^{-1}$

$r = ?$

Now,

the bond length of CO,

$$r_{CO} = \sqrt{\frac{I}{\mu}}$$

$$= \sqrt{\frac{14.57 \times 10^{-47}}{11.38 \times 10^{-27}}}$$

$$= 1.13 \times 10^{-10}\ m$$

$$= 11.3 \times 10^{-11}\ m. \quad \text{(Ans)}$$

$$= 0.113\ nm.$$

$$I = \frac{h}{8\pi^2 Bc}$$

$$= \frac{6.626 \times 10^{-34}}{8 \times \pi^2 \times 1.92 \times 3 \times 10^{10}}$$

$$= 1.4 \times 10^{-46}\ kgm^2$$

$$= 14.57 \times 10^{-47}\ kgm^2$$

07. In the rotational spectrum of CO at 25°C, the $J = 3 \rightarrow J = 4$ transition is more intense than the $J = 0 \rightarrow J = 1$ transition. Explain. [2008]

➪

(2012, 2011) 08. The rotational spectrum of $^{79}B^{19}F$ shows a series of equidistance lines spaced $0.71433\ cm^{-1}$ apart. Calculate (i) Rotational constant (ii) The moment of inertia (iii) The bond length of the molecule.

[given atomic mass: $^{79}B = 131.03 \times 10^{-27}$ kg

$^{19}F = 31.55 \times 10^{-27}$ kg.

$\hbar = 6.626 \times 10^{-34}$ Js]

[2005]

(i) Here, $2B = 0.71433\ cm^{-1}$.

So, rotational constant, $B = 0.357\ cm^{-1}$.

(ii) $I = \dfrac{\hbar}{8\pi^2 BC}$

$= \dfrac{6.626 \times 10^{-34}}{8 \times \pi^2 \times 0.357 \times 3 \times 10^{10}}$

$= 7.84 \times 10^{-46}\ kgm^2$.

(iii) $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{131.03 \times 10^{-27} \times 31.55 \times 10^{-27}}{131.03 \times 10^{-27} + 31.55 \times 10^{-27}} =$

$= 25.43 \times 10^{-27}$ kg.

P.T.O.

Again, $r = \sqrt{\frac{I}{\mu}}$

$= \sqrt{\frac{7.84 \times 10^{-46}}{25.43 \times 10^{-27}}}$

$= 1.756 \times 10^{-10} m.$

09. The inter-nuclear distance for the molecule NO is 1.51 Å. Calculate the energies for $J=2$ & $J=3$ & find out at which wave number transition $J=2 \longrightarrow J=3$ will occur? [2004]

⇨

Extra Maths

2013

10. The rotational constant for $H^{35}Cl$ is observed to be 10.5909 cm^{-1}. What are the values of B' for $H^{37}Cl$ & $^2D^{35}Cl$? [Exercise 2.3]

➪ Here, for $H^{37}Cl$

for $(H^{35}Cl) \rightarrow \mu = \frac{35 \times 1}{35+1} = 0.972$ g.

for $(H^{37}Cl) \rightarrow \mu' = \frac{37 \times 1}{37+1} = 0.974$ g.

Note: reduced mass এর একক amu তে করলে; kg তে নেয়া লাগে।

$B = 10.5909\ cm^{-1}$

We know/We get, (for $H^{37}Cl$)

$$\frac{B}{B'} = \frac{\mu'}{\mu}$$ [How? see Math no 02 (Prev. Yr)]

$$\Rightarrow B' = \frac{B\mu}{\mu'}$$

$$\Rightarrow B' = \frac{10.5909 \times 0.972}{0.974}$$

$$\therefore B' = 10.569\ cm^{-1}.$$ (Ans)

Again, for, $^2D^{35}Cl$

We know (for $^2D^{35}Cl$)

$$\frac{B}{B'} = \frac{\mu'}{\mu}$$

$$\Rightarrow B' = \frac{B.\mu}{\mu'} = \frac{10.5909 \times 0.972}{1.892}$$

$$= 5.441\ cm^{-1}$$ (Ans)

Here,

for $(H^{35}Cl)$, $\mu = \frac{35 \times 1}{35+1} = 0.972$

for $(^2D^{35}Cl)$, $\mu' = \frac{2 \times 35}{2+35} = 1.892$ g

$B = 10.5909\ cm^{-1}$

11. The diatomic molecule HCl has B value of $10.593\ cm^{-1}$ & a centrifugal distortion constant $D = 5.3 \times 10^{-4}\ cm^{-1}$. Estimate the vibrational frequency of the molecule & force constant. [Ex. 2.7]

TU-11, 2012

➪ We know,

$$\bar{\omega}^2 = \frac{4B^3}{D}$$

$$= \frac{4 \times (10.593)^3}{5.3 \times 10^{-4}}$$

$$= 8.97 \times 10^6\ cm^{-2}$$

$$\therefore \bar{\omega} = 2.995 \times 10^3\ cm^{-1}.$$

μ_{HCl} $m_1 = 1.008 \times 1\ amu$

$= 1.008 \times 1.66 \times 10^{-27}\ kg$

$= 1.673 \times 10^{-27}\ kg.$

$m_2 = 35.5 \times 1.66 \times 10^{-27}$

$= 58.93 \times 10^{-27}\ kg.$

$\therefore \mu = \frac{m_1 m_2}{m_1 + m_2} = 1.627 \times 10^{-}$

& Given, $\bar{\omega}_{obs} = 2995\ cm^{-1}$

Now, force constant $f = 4\pi^2 \bar{\omega}_{obs}^2 c^2 \mu$

$= 4 \times \pi^2 \times (2995)^2 \times (3 \times 10^{10})^2 \times 1.627 \times 10^{-}$

$= 518.5\ Nm^{-1}.$

2012. The equilibrium oscillation frequency for NO is $1904\ cm^{-1}$. Calculate the force constant of the molecule.

➪ We know, force constant,

$$K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu_{NO}$$

$$= 4 \times \pi^2 \times (1904)^2 \times (3 \times 10^{10})^2 \times 1.24 \times 10^{-26}$$

$$= 1.597 \times 10^3\ kg\ s^{-2}$$

$$= 1.597 \times 10^3\ Nm^{-1}.$$

Here,

$\bar{\omega}_e = 1904\ cm^{-1}$

$c = 3 \times 10^{10}\ cms^{-1}.$

$\mu_{NO} = \frac{14 \times 1.66 \times 10^{-27} \times 16 \times 1.66 \times 10^{-2}}{14 \times 1.66 \times 10^{-27} + 16 \times 1.66 \times 10^{-2}}$

$= 1.24 \times 10^{-26}\ kg$

Lecture 8 & 9 on ν̄ and I

13. Calculate the moment of inertia of CO, where the rotational spectrum for 1st line (J=0) is 3.84 cm⁻¹

⇨ Now,

$$\bar{\nu}_{J=0 \to J=1} = \varepsilon_{J=1} - \varepsilon_{J=0}$$

$$= B[1(1+1) - 0.(0+1)]$$

$$= 2B.$$

Here,
$2B = 3.84235\ cm^{-1}$
$\therefore B = 1.92118\ cm^{-1}$
$c = 3 \times 10^{10}\ cms^{-1}$
$h = 6.626 \times 10^{-34}\ Js.$

So, the moment of inertia,

$$I = \frac{h}{8\pi^2 Bc}$$

$$= \frac{6.626 \times 10^{-34}}{8 \times \pi^2 \times 1.92118 \times 3 \times 10^{10}}$$

$$= 14.56 \times 10^{-47}\ kgm^2. \quad \text{(Ans)}$$

$\mu_{CO} = \frac{m_C m_O}{m_C + m_O}$

$I = \mu r^2$

$\therefore r = \sqrt{\frac{I}{\mu}}$

14. The rotational spectrum of $^{79}Br^{19}F$ shows a series of equidistant lines 0.71433 apart. Calculate — Tu-11

(I) rotational constant, B, (II) moment of inertia, (III) bond length of the molecule (IV) Determine the wave num of the J=9 → J=10 transition. (V) find which transition gives rise to the most intense spectral line at room temp. (say 300K) (VI) Calculate the number of revolutions per second which the Br-F molecule undergoes when in (a) the J=0 state, (b) the J=1 state & (c) the J=10 state.

[Ex. 2.2]

P.T.O.